Aluvextra Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-932/A

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process parameters identified during development. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Deviations observed during the campaign were investigated and closed with no impact to product quality. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. No changes to the approved analytical procedures are proposed in this submission.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

P.4 Control of Excipients

For the commercial formulation F-932/A, Polyplasdone XL-10 (Crospovidone, disintegrant) is used.

The approach described herein is consistent with FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms and Ph. Eur. general monograph 2034.

Basis for Selection

Polyplasdone XL-10 was selected as the commercial grade of crospovidone on the basis of improved blend uniformity in geometric dilution studies.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-2728 Rev 02) lists Crospovidone as Polyplasdone XL-10.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Deviations observed during the campaign were investigated and closed with no impact to product quality. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process parameters identified during development.